Real Time Driver Drowsiness Detection System

Tushar(221030228), Piyush Sharma(221030235), Parth Agarwal(221030304)
Department of Computer Science & Engineering and Information Technology
Under supervision of **Dr. Monika (Associate Professor SG)**
Jaypee University of Information Technology, Waknaghat, Solan - 173234 (India)

PROBLEM STATEMENT

Road accidents due to driver drowsiness are a major concern especially in the case of long and repetitive driving conditions. Designing an effective real-time drowsiness detection system is still a challenging task due to varying lighting conditions, camera angles, head movements, facial obstructions and user to user differences. In addition, many deep learning models require high computational resources, which makes it hard to deploy them on mobile devices. This requires a lightweight and robust vision-based system that can run efficiently on a standard smartphone and remain stable in real-time performance under practical driving conditions.

OBJECTIVE

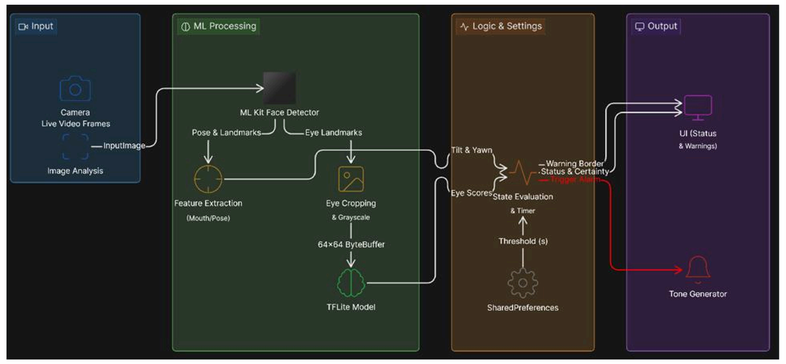
- To perform an exhaustive research analysis of various existing methods of software-based detection of drowsiness with a focus on computer vision as well as deep learning methodologies to understand how these solutions work, their benefits, and
- To design a detection system that will allow processing live video input in real-time, performing face detection, and obtaining facial landmarks relevant to driver's drowsiness
- To create a model using a Convolutional Neural Network (CNN) that will allow classifying driver's drowsiness levels, including alertness, closed eyes, and yawning. Implement additional functionality related to behaviour analysis, namely detection of the presence of closed eyes, and head \
- To convert and optimize the designed model for use in TensorFlow Lite (TFLite) format for to be used on mobiles and develop an Android mobile application using Kotlin programming language to run inference from the camera and utilize the pre-trained model

METHODOLOGY

The proposed system is an integration of computer vision, facial landmark detection and deep learning techniques for real-time drowsiness detection on Android devices. We are using CameraX for capturing live video frames and Google ML Kit for face tracking and landmark extraction. In the development of the model, we used Dlib framework based facial landmark based preprocessing techniques to identify important facial regions, such as the eyes and head orientation.

The eye and mouth regions are extracted from the detected faces, transformed to greyscale images, resized to a fixed size and normalised before being fed into a lightweight Convolutional Neural Network (CNN) model converted to TensorFlow Lite (TFLite) format. The CNN model was trained using labelled eye image datasets to classify the eye states to be open and closed.

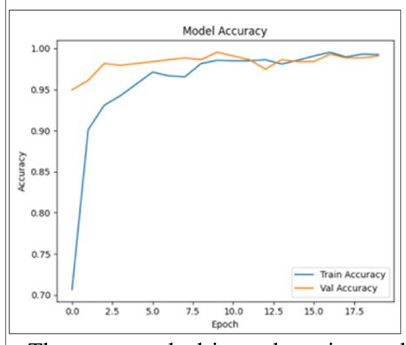
Workflow:

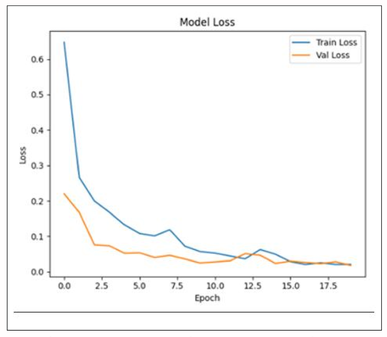


IMPLEMENTATION DETAILS

The system is a mobile-based driver drowsiness detection application composed of multiple integrated modules. It uses Python-based preprocessing techniques to prepare eye and yawning datasets, while a lightweight CNN model built with TensorFlow and Keras classifies eye states as open or closed. The trained model is converted to TensorFlow Lite for efficient real-time inference on Android devices. Using CameraX and Google ML Kit, the application detects facial landmarks, monitors eye closure and head tilt, and triggers audio-visual alerts when signs of drowsiness are detected. Additionally, the app integrates MapLibre SDK and Geoapify APIs to provide route calculation and trip monitoring features within the same platform.

RESULTS





The proposed driver drowsiness detection system demonstrated strong performance during both training and validation phases. The CNN-based eye-state classification and yawning detection model achieved an overall accuracy of approximately 96%, indicating reliable classification of open and closed eye states. The training process showed consistent convergence with decreasing loss values and stable validation accuracy, confirming effective learning and reduced overfitting.

CONCLUSION & FUTURE SCOPE

Conclusion:

- Developed a real-time driver drowsiness detection system using deep learning, computer vision, and Android-based mobile deployment. The system successfully detects prolonged eye closure , yawning and head tilt while generating visual and audio
- Successfully trained and deployed a lightweight CNN model with approximately **96% accuracy** for eye-state classification using TensorFlow, Keras, and TensorFlow
- The project objectives were achieved successfully, and the system demonstrated effective real-time performance on consumer-grade mobile devices without requiring specialized hardware.

Future Scope and Enhancements:

- Future improvements may include integrating steering behavior analysis and advanced driver behavior monitoring techniques to further improve the accuracy and reliability of drowsiness detection under real-world driving
- The system can be enhanced using cloud and IoT technologies for fleet-level monitoring, remote emergency alerts, trip analytics, and synchronization of driver safety data across multiple devices and
- In the future, such AI-based driver monitoring systems could become a standard built-in safety feature in vehicles, where every car comes pre-installed with real-time drowsiness detection and alert mechanisms to improve overall road safety.

REFERENCES

1. T. Gautam et al., “Novel CNN based model using LSTM for driver drowsiness detection” (2025)
2. S. K. Ghanta et al., “VigilEye: YOLOv5 Vision Transformer for Drowsiness Detection” (2022)
3. V. K. Tran et al., “Spatial Self-Attention + LSTM + Federated Learning for Privacy-Preserving Drowsiness Detection” (2025)
4. Chongqing Jiaotong Univ., “A Real-Time Driver Drowsiness Detection Method by integrating lightweight YOLOv5s and facial 3D Keypoints” (2023)